# DMVPN Fase 3 IKEv2

VPN Hub and Spoke con DMVPN Fase 3 y Enrutamiento Dinámico

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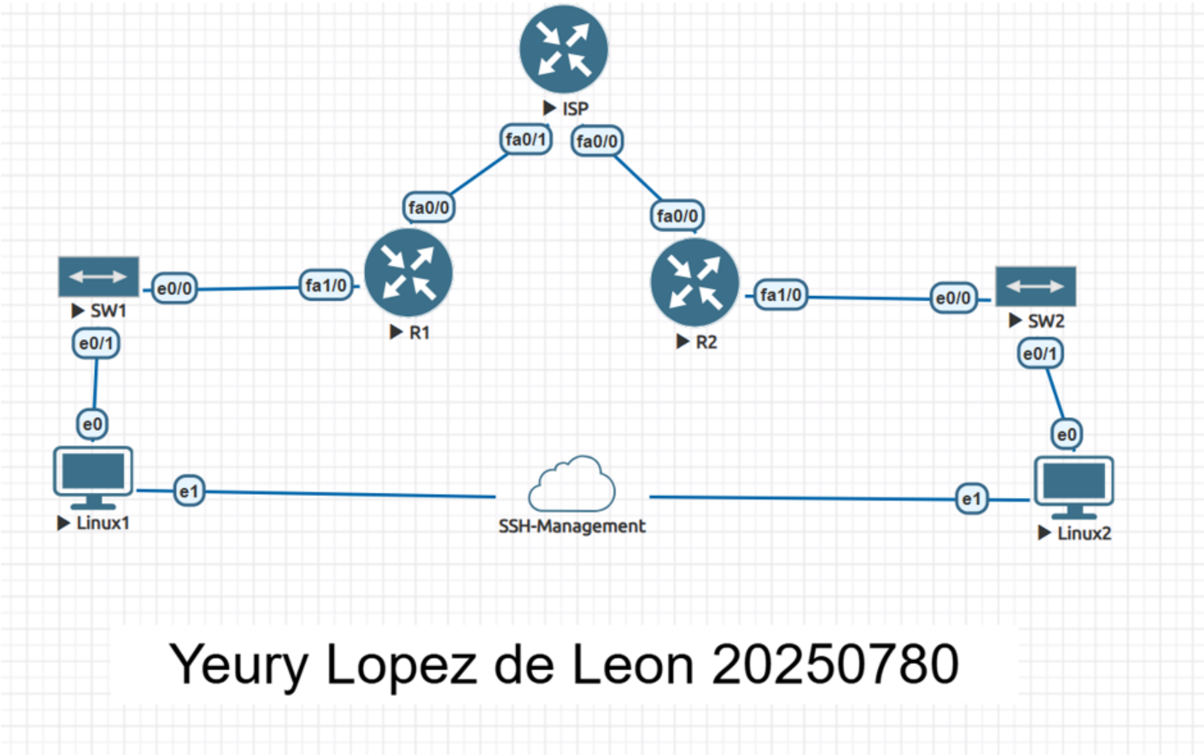
# 1. Objetivo

Configurar una VPN Hub and Spoke punto a multipunto DMVPN Fase 3 con IKEv2 y enrutamiento dinámico OSPF. En la Fase 3, el Hub usa ip nhrp redirect para indicar a los Spokes una ruta más óptima, y los Spokes usan ip nhrp shortcut para instalar esa ruta directamente en su tabla de enrutamiento. IKEv2 reemplaza a IKEv1 como mecanismo de negociación.

## Topología de Red

La topología DMVPN consiste en un router HUB central y dos Spokes conectados a través de un router ISP. Cada router tiene una LAN con un host Linux para demostrar la conectividad.




## Direccionamiento IP

| Dispositivo | Interfaz | Dirección IP | Rol |
|---|---|---|---|
| HUB | Fa0/0 | 192.168.7.1/24 | LAN |
| HUB | Fa1/0 | 10.7.82.1/30 | WAN → ISP |
| ISP | Fa0/0 | 10.7.82.2/30 | WAN → HUB |
| ISP | Fa3/0 | 10.7.83.1/30 | WAN → SPOKE1 |
| ISP | Fa1/0 | 10.7.84.1/30 | WAN → SPOKE2 |
| SPOKE1 | Fa0/0 | 10.7.83.2/30 | WAN → ISP |
| SPOKE1 | Fa1/0 | 192.168.80.1/24 | LAN |
| SPOKE2 | Fa0/0 | 10.7.84.2/30 | WAN → ISP |
| SPOKE2 | Fa1/0 | 192.168.78.1/24 | LAN |
| HUB Tunnel0 | — | 172.16.78.1/24 | DMVPN |
| SPOKE1 Tunnel0 | — | 172.16.78.2/24 | DMVPN |
| SPOKE2 Tunnel0 | — | 172.16.78.3/24 | DMVPN |

# 2. Parámetros VPN

| Parámetro | Valor |
|---|---|
| Tipo de VPN | DMVPN Fase 3 con IKEv2 |
| Protocolo Túnel | mGRE (Multipoint GRE) |
| Protocolo NHRP | NHRP Network-ID 780 |
| Comando Hub | ip nhrp redirect |
| Comando Spoke | ip nhrp shortcut |
| Enrutamiento Dinámico | OSPF Proceso 1 — Area 0 |
| IKEv2 — Cifrado | AES-CBC-128 |
| IKEv2 — Integridad | SHA-1 |
| IKEv2 — Autenticación | Pre-Shared Key (PSK) |
| Fase 2 — Modo | Transport |
| Pre-Shared Key | Yeury0780 |

# 3. Configuración

## 3.1 Configuración HUB

```
crypto ikev2 proposal PROP-IKEv2
 encryption aes-cbc-128
 integrity sha1
 group 2
!
crypto ikev2 policy POL-IKEv2
 proposal PROP-IKEv2
!
crypto ikev2 keyring KR-IKEv2
 peer SPOKES
  address 0.0.0.0 0.0.0.0
  pre-shared-key Yeury0780
  !
!
!
crypto ikev2 profile PROF-IKEv2
 match identity remote address 0.0.0.0
 authentication remote pre-share
 authentication local pre-share
 keyring local KR-IKEv2
```

```
interface Tunnel0  ip address 172.16.78.1 255.255.255.0  no ip redirects  ip nhrp map multicast dynamic  ip nhrp network-id 780  ip nhrp redirect  tunnel source FastEthernet1/0  tunnel mode gre multipoint  tunnel protection ipsec profile DMVPN-PROFILE  ip ospf network point-to-multipoint  ip ospf priority 255
```

## 3.2 Configuración SPOKE1

```
interface Tunnel0  ip address 172.16.78.2 255.255.255.0  ip nhrp map 172.16.78.1 
10.7.82.1  ip nhrp map multicast 10.7.82.1  ip nhrp network-id 780  ip nhrp nhs 
172.16.78.1  ip nhrp shortcut  tunnel source FastEthernet0/0  tunnel mode gre 
multipoint  tunnel protection ipsec profile DMVPN-PROFILE  ip ospf network point-
to-multipoint  ip ospf priority 0
```

# 4. Verificación y Funcionamiento

## 4.1 Estado DMVPN

```
HUB#show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
     1 10.7.83.2           172.16.78.2    UP 00:12:48     D
     1 10.7.84.2           172.16.78.3    UP 00:12:32     D
```

Los Spokes registrados en estado UP confirman que el DMVPN Fase 3 está activo.

## 4.2 Estado IKEv2 SA

```
HUB#show crypto ikev2 sa
 IPv4 Crypto IKEv2  SA

Tunnel-id Local                 Remote                fvrf/ivrf            Status
2         10.7.82.1/500         10.7.84.2/500         none/none            READY
      Encr: AES-CBC, keysize: 128, Hash: SHA96, DH Grp:2, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/780 sec

Tunnel-id Local                 Remote                fvrf/ivrf            Status
1         10.7.82.1/500         10.7.83.2/500         none/none            READY
      Encr: AES-CBC, keysize: 128, Hash: SHA96, DH Grp:2, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/797 sec

 IPv6 Crypto IKEv2  SA
```

Las sesiones IKEv2 en estado READY confirman que el cifrado IKEv2 está activo entre el HUB y los Spokes.

## 4.3 Estado OSPF

```
HUB#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.78.1      0   FULL/  -        00:01:55    172.16.78.3     Tunnel0
192.168.80.1      0   FULL/  -        00:01:32    172.16.78.2     Tunnel0
```

Los vecinos OSPF en estado FULL confirman el enrutamiento dinámico activo.

## 4.4 Tabla de rutas

```
HUB#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 10.7.82.2 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.7.82.2
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.7.82.0/30 is directly connected, FastEthernet1/0
L        10.7.82.1/32 is directly connected, FastEthernet1/0
      172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C        172.16.78.0/24 is directly connected, Tunnel0
L        172.16.78.1/32 is directly connected, Tunnel0
```

Las rutas OSPF confirman la distribución correcta de rutas entre todos los sitios.

## 4.5 Demostración de conectividad

```
root@ubuntu22-server:~# ping -c 4 192.168.78.2
PING 192.168.78.2 (192.168.78.2) 56(84) bytes of data.
64 bytes from 192.168.78.2: icmp_seq=1 ttl=61 time=118 ms
64 bytes from 192.168.78.2: icmp_seq=2 ttl=61 time=271 ms
64 bytes from 192.168.78.2: icmp_seq=3 ttl=61 time=545 ms
64 bytes from 192.168.78.2: icmp_seq=4 ttl=61 time=812 ms
```

El ping exitoso demuestra la conectividad completa entre todos los sitios con DMVPN Fase 3 e IKEv2.